

Online Resource 1

Lightweight Open-Set Reliability Calibration for Frozen Open-Vocabulary Object Detectors

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This document supports the main paper. Section S1 gives method details that were shortened in the paper. Section S2 describes the detector settings, class vocabularies and computing environment. Section S3 reports the additional results referred to in Sections 5–7 of the paper. Code, cached detector outputs, fitted models and image splits are provided in Online Resource 2.

Notation. Unless stated otherwise, results refer to YOLO-World-l on the 4000 COCO test images, with matching at IoU 0.50.

Term	Meaning
Raw	Detector confidence with one global threshold chosen on calibration images
SC	Ranker using confidence and predicted class, with class-specific confidence coefficients
SG	Ranker using confidence and box geometry, without class terms
SCG	Full ranker using confidence, predicted class and geometry, with class-specific coefficients
RF	Mode that selects the score function and threshold with the highest calibration B
AP-C	RF restricted to policies meeting the calibration AP and AP50 tolerances relative to Raw (Table 6 of the paper)
P, R	Precision and recall for known classes
UFA	Number of annotated unknown objects covered by at least one accepted false detection
BG FP	Number of accepted detections with IoU below the matching threshold for every annotated object
URR	Unknown reject rate, $1 - \text{UFA} / (\text{number of annotated unknown objects})$
B	Harmonic mean of P, R and URR
Full-list AP	AP computed on all candidate detections before the acceptance threshold
Accepted-set AP	AP computed only on the detections that pass the threshold

S1 Method details

S1.1 Weighted logistic objective

The ranker in Eq. (3) of the paper is fitted by minimizing

$$L(\tilde{\theta}) = 1/2 \|\tilde{\theta}\|^2 + C \sum_{i \in I_{fit}} a_{q_i} v_i l(q_i, r_i), l(q, r) = -q \log r - (1 - q) \log(1 - r), \quad (S1)$$

where $\tilde{\theta} = (\theta_0, \theta)$ contains the intercept and the feature coefficients, I_{fit} is the set of fitting candidates, $q_i \in \{0, 1\}$ indicates a known true positive, and r_i is the ranker output. The risk weight v_i is 2.0 for negatives that overlap an annotated unknown object at IoU 0.50, 1.25 for negatives whose IoU is below 0.50 with every annotation, and 1.0 for all other candidates. The class weight a_q balances positives and negatives and is computed from their risk-weighted counts. As in the liblinear solver, the intercept is penalized together with the other coefficients. The final model uses $C = 0.1$ (Section S1.5).

S1.2 Score functions and policy grid

A policy consists of a score function g and a threshold τ ; a candidate is accepted when $g(d_i) \geq \tau$. Let s_i be the raw detector confidence, r_i the ranker output and ρ_y the class prior defined in Section S1.3. Five families of score functions are used:

$$\begin{aligned} \text{raw: } g &= s_i; \text{direct: } g = r_i; \text{blend: } g = (1 - \alpha)s_i + \alpha r_i; \\ \text{class scaling: } g &= s_i \rho_y^\gamma; \text{light gate: } g = s_i \text{clip}((1 - \beta) + \beta r_i, 0, 1). \end{aligned} \quad (\text{S2})$$

Class scaling and raw confidence keep the order of candidates within each class; the other families can change it. The light-gate multiplier lies between $1 - \beta$ and 1.

Table S1. Score functions and thresholds searched during policy selection.

Family	Parameter values	Number of score functions	Keeps within-class order
Raw confidence	—	1	Yes
Direct ranker score	—	1	No
Blend	$\alpha \in \{0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1\}$	7	No
Class scaling	$\gamma \in \{0.25, 0.5, 0.75, 1\}$	4	Yes
Light gate	$\beta \in \{0.05, 0.1, 0.2\}$	3	No
Total		16 (15 distinct)	

Note. The blend with $\alpha = 1$ is identical to the direct score. For every score function, the threshold set contains 13 fixed values (0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90), the 25th, 50th, 70th, 80th, 90th and 95th percentiles of the scores on the policy-selection images, and the maximum score. On the primary calibration set, this gives 320 policies, of which 300 are distinct; coincident threshold values are deduplicated. The same grid is searched by RF and AP-C. SCG searches only the thresholds of the direct score.

S1.3 Class priors

For predicted class y , let N_y be the number of fitting candidates, and let TP_y , UNK_y and BG_y be the numbers of these candidates that are known true positives, are non-TP detections overlapping an annotated unknown object at IoU 0.50, or have IoU below 0.50 with every annotation. With add-one smoothing,

$$p_{TP,y} = \frac{TP_y + 1}{N_y + 2}, p_{nonUNK,y} = \frac{N_y - UNK_y + 1}{N_y + 2}, p_{nonBG,y} = \frac{N_y - BG_y + 1}{N_y + 2}, \quad (\text{S3a})$$

$$\rho_y^0 = (p_{TP,y} p_{nonUNK,y} p_{nonBG,y})^{1/3}, \rho_y = \text{clip}\left(\frac{\rho_y^0}{\max_{y'} \rho_{y'}^0}, 0.20, 1\right). \quad (\text{S3b})$$

Because $\rho_y \leq 1$, class scaling can only lower a score. When only known-class labels are available (Section S3.4), $\rho_y^0 = p_{TP,y}$.

S1.4 Folding the class interactions into a lookup table

In Eq. (2b) of the paper, $\mathbf{z}_i$ contains the standardized numeric vector $\tilde{\mathbf{n}}_i$ (confidence, its logit and the eight geometry terms), the one-hot class vector $\mathbf{c}_{y_i}$, and their product. The linear part of the ranker can therefore be regrouped by class:

$$r_i = \sigma \left(\mathbf{b}_{y_i} + \omega_{y_i}^T \tilde{\mathbf{n}}_i \right), \omega_y = \theta_{num} + \Theta_y, \mathbf{b}_y = \theta_0 + \theta_{cls,y}, \quad (\text{S4})$$

where θ_{num} are the coefficients of $\tilde{\mathbf{n}}_i$, Θ_y are the ten interaction coefficients of class y , and $\theta_{cls,y}$ is the coefficient of the class indicator. At deployment, the ranker therefore needs one ten-dimensional coefficient vector and one offset per class. The folded and unfolded scores differ by at most 6.1×10^{-16} on all 100,497 test candidates, and they accept identical detections. A class that does not occur in the fitting data uses θ_{num} and θ_0 only.

S1.5 Ranker families and the choice of C

Four ranker families were compared:

- **Additive** (30 features): the ten numeric terms and the class indicators.
- **Class interactions** (230 features): the additive features plus the product of every numeric term with every class indicator. This is the final SCG ranker.
- **Class profiles** (39 features): the additive features plus nine profile features. For each known class, the mean and spread of the eight geometry terms are estimated from known true positives in the fitting images. The profile features are the eight absolute standardized deviations from this profile and their Euclidean norm. Class means and second moments are shrunk toward the pooled known-TP profile with a weight of 20 observations. Each standard deviation is bounded below by 0.1 times the pooled standard deviation and by 10^{-6} . The nine derived features are standardized on fitting candidates.
- **Class interactions with profiles** (239 features).

Each family was fitted with $C \in \{0.1, 0.5, 1, 2\}$ on three random 500/500 divisions of the 1000 calibration images: one half was used for fitting and the other for policy selection. Table S2 reports the mean B on the selection halves. Class interactions with $C = 0.1$ had the highest mean and were fixed before any test evaluation. The matched SC control for each configuration removes all geometry terms. All 16 configurations lie within 0.0014 of each other, so calibration B separates them only weakly. The nested experiment in Section S3.6 shows that other families can be selected when less calibration data are available.

Table S2. Mean selection B of the 16 ranker configurations on three internal calibration splits.

Family	C = 0.1	C = 0.5	C = 1	C = 2
Additive	0.7810	0.7808	0.7810	0.7810
Class interactions	0.7815	0.7807	0.7807	0.7804
Class profiles	0.7804	0.7805	0.7805	0.7805
Class interactions with profiles	0.7809	0.7805	0.7805	0.7801

Note. Each value is the mean over three 500/500 calibration splits. The splits overlap, so the values are not independent replications. Bold marks the selected configuration.

S2 Experimental settings

S2.1 Detector and inference settings

Table S3 lists the settings used to generate the candidate detections. Both detectors were used with frozen weights. After the detector's own post-processing, the same output-side suppression was applied to every stream: within each image and predicted class, a lower-scoring box is removed if its IoU with a higher-scoring box is at least 0.50, or if at least 90% of its area lies inside that box.

To check that these settings are complete, we reran YOLO-World-l on all 5000 COCO images with the 20-class prompt of the primary protocol. The rerun reproduced the Raw test operating point of the primary stream exactly (13,283 accepted detections, UFA 299, BG FP 2875); full-list AP was 0.4860, compared with 0.4859.

Table S3. Detector and inference settings.

Setting	YOLO-World-l	Grounding DINO tiny
Model	yolov8l-worldv2.pt (Ultralytics)	IDEA-Research/grounding-dino-tiny (Hugging Face)
Prompt	Known class names in increasing COCO category ID	Known class names separated by periods, e.g. "person. bicycle. car."
Text encoder	CLIP ViT-B/32	Built-in
Input size	960 pixels; aspect ratio kept, padded to the model stride	Shortest side 800, longest side at most 1333
Image normalization	Ultralytics default	RGB mean (0.485, 0.456, 0.406), standard deviation (0.229, 0.224, 0.225)
Score threshold	0.001	Box 0.05, text 0.20
Detector NMS	Class-aware, IoU 0.70	No additional NMS in the Hugging Face box/text post-processor
Maximum detections per image	300	No limit
Numerical precision, batch size	FP32, 1, no test-time augmentation	FP32, 1, no test-time augmentation
Output-side suppression	Class-wise IoU 0.50 and containment 0.90 (see text)	Same
Box coordinates	Original image pixels	Original image pixels

Note. The LVIS-Clear-Mini-300 stream uses the YOLO-World-l settings with its four-class prompt. Images with no remaining candidate are kept in all counts. Candidate totals: COCO / YOLO-World-l 125,428 (24,931 calibration, 100,497 test); COCO / Grounding DINO 35,488 (7,248 calibration, 28,240 test); LVIS / YOLO-World-l 9,096 (3,086 calibration, 6,010 test).

S2.2 Software and hardware

Experiments used Python 3.11, NumPy 2.0.1, pandas 3.0.2, scikit-learn 1.8.0, PyTorch 2.5.1 with CUDA 12.1, Ultralytics 8.4.39, Hugging Face transformers 5.7.0 and pycocotools 2.0.10. Detector inference ran on an NVIDIA RTX 4070 Laptop GPU (8 GB). Timing measurements used one thread of an Intel Core i7-12800HX CPU. Exact package versions and file checksums are listed in Online Resource 2.

S2.3 Image splits and class vocabularies

The 5000 COCO val2017 images were shuffled with random seed 1201; the first 1000 form the calibration set and the remaining 4000 the test set. LVIS-Clear-Mini-300 uses 105 calibration and 195 test images. All image identifiers are given in Online Resource 2.

The primary COCO protocol treats the first 20 COCO categories (by category ID) as known and the remaining 60 as annotated unknown objects. This set is not the PASCAL VOC class list; VOC20 is evaluated separately in Section S3.5. Table S4 lists the known classes of all vocabularies. In each COCO vocabulary, all categories not listed as known are treated as annotated unknown objects.

Table S4. Known classes in each vocabulary.

Vocabulary	Known classes (20 for COCO, 4 for LVIS)	Known / unknown test objects
COCO, primary (first 20 categories)	person, bicycle, car, motorcycle, airplane, bus, train, truck, boat, traffic light, fire hydrant, stop sign, parking meter, bench, bird, cat, dog, horse, sheep, cow	14,775 / 14,613
COCO, VOC20	person, bicycle, car, motorcycle, airplane, bus, train, boat, bird, cat, dog, horse, sheep, cow, bottle, chair, couch, potted plant, dining table, tv	16,912 / 12,476
COCO, random set 1 (seed 611)	car, motorcycle, airplane, bus, horse, bear, zebra, handbag, frisbee, sports ball, skateboard, spoon, orange, carrot, hot dog, couch, mouse, remote, microwave, vase	5,229 / 24,159
COCO, random set 2 (seed 612)	airplane, bus, traffic light, bench, cat, horse, cow, suitcase, skis, skateboard, cup, fork, hot dog, mouse, keyboard, toaster, sink, book, clock, vase	5,183 / 24,205
COCO, random set 3 (seed 613)	car, train, truck, bear, snowboard, banana, sandwich, orange, broccoli, carrot, hot dog, pizza, donut, toilet, laptop, remote, microwave, oven, teddy bear, hair drier	4,728 / 24,660
LVIS-Clear-Mini-300	bottle, cup, chair, bowl. Unknown: apple, banana, orange fruit, plate, fork, vase, glass (drink container), knife	218 / 207

Note. Prompts list the known class names in increasing category ID. The three random sets were drawn once with the stated seeds before any detector was run on them.

S3 Additional results

S3.1 Paired uncertainty for the main comparisons

Table S5 gives 95% intervals for the two comparisons shown in Fig. 7(a) of the paper. The intervals come from paired resampling of test images: all detections and annotations of an image are resampled together, and the fitted models and thresholds are kept fixed. RF improves B and reduces UFA relative to Raw, with no detectable change in AP. Adding geometry to the matched SC model reduces UFA and lowers AP, but its B interval includes zero.

Table S5. Paired differences with 95% image-bootstrap intervals (COCO, YOLO-World-l).

Metric	RF/AP-C – Raw	SCG – SC
B	+0.0060 [0.0038, 0.0083]	+0.0001 [−0.0021, 0.0022]
P	+0.0103 [0.0063, 0.0146]	+0.0004 [−0.0036, 0.0044]
R	+0.0012 [−0.0020, 0.0044]	−0.0007 [−0.0039, 0.0023]
UFA	−123 [−149, −98]	−24 [−40, −8]
BG FP	−47 [−107, 12]	+15 [−54, 78]
Full-list AP	−0.0002 [−0.0024, 0.0024]	−0.0057 [−0.0087, −0.0011]
Full-list AP50	+0.0006 [−0.0024, 0.0033]	−0.0028 [−0.0068, 0.0017]

Metric	RF/AP-C – Raw	SCG – SC
Full-list AP75	+0.0002 [−0.0024, 0.0031]	−0.0048 [−0.0085, 0.0004]

Note. Intervals for B, P, R, UFA and BG FP use 1000 resamples; AP intervals use 500 resamples, because each resample requires a full AP evaluation. RF and AP-C select the same policy, so one column covers both. Two further comparisons on the same images: RF with the final ranker versus RF with the additive ranker, $\Delta B = +0.0002$ [−0.0013, 0.0019]; RF versus SC with the same 16-function policy search, $\Delta B = -0.0001$ [−0.0022, 0.0017].

S3.2 Contribution of geometry

This section separates the effect of geometry from the rest of the framework. In all comparisons, SC and SCG share the class-specific confidence terms, labels, weights, regularization and threshold rules; they differ only in the geometry terms and their class interactions.

Across detectors and protocols (Table S6). On the primary YOLO-World stream, geometry shows no clear B improvement, reduces UFA by 24 objects (14%) and lowers full-list AP by 0.0057. On Grounding DINO the effect is much larger. Class terms alone raise B over Raw by only 0.0018 (0.7666 $\rightarrow$ 0.7684). Adding geometry raises it by a further 0.0094 ($\rightarrow$ 0.7778), removes 1753 background accepts (40%) and raises precision by 8.2 percentage points, at a recall cost of 5.0 points. Geometry therefore provides most of the improvement on this detector. On the small LVIS protocol, all reported geometry intervals include zero.

Table S6. Effect of adding geometry to the matched SC model (SCG – SC), with 95% intervals.

Detector / protocol	SC $\rightarrow$ SCG: B	ΔB	ΔUFA	$\Delta BG\ FP$	ΔP	ΔR	ΔAP
YOLO-World-l / COCO	0.7791 $\rightarrow$ 0.7792	+0.0001 [−0.0021, 0.0022]	−24 [−40, −8]	+15 [−54, 78]	+0.0004	−0.0007	−0.0057 [−0.0087, −0.0011]
Grounding DINO tiny / COCO	0.7684 $\rightarrow$ 0.7778	+0.0094 [0.0044, 0.0149]	−24 [−40, −9]	−1753 [−2079, −1460]	+0.0817	−0.0499	+0.0034
YOLO-World-l / LVIS-Mini	0.5927 $\rightarrow$ 0.6140	+0.0213 [−0.0219, 0.0609]	+5 [−3, 13]	−13 [−32, 5]	+0.0260	+0.0275	+0.0020

Note. Raw B is 0.7729 (YOLO-World / COCO), 0.7666 (Grounding DINO / COCO) and 0.5480 (LVIS). AP intervals are available for the primary stream only. When SC and SCG each search the full 16-function policy grid, the B differences are −0.0001 [−0.0022, 0.0017] (YOLO-World / COCO), +0.0114 [0.0076, 0.0152] (Grounding DINO) and +0.0194 [−0.0213, 0.0583] (LVIS).

Across repeated splits (Table S7). In five further random calibration/test splits of the COCO images, RF selected the direct ranker at threshold 0.80 every time, so SCG, RF and AP-C are the same policy in each split. The geometry effect on B changes sign (two positive, three negative; mean −0.0003). The UFA reduction is smaller than on the primary split (mean −4.2), whereas the AP loss is consistent (−0.0066 to −0.0099). RF improves B over Raw in all five splits (+0.0029 to +0.0090).

Table S7. Raw, SC and SCG (= RF = AP-C) on five repeated calibration/test splits.

Split (seed)	Raw B	SC B	SCG B	ΔB (SCG – SC)	ΔUFA	$\Delta BG\ FP$	ΔAP
1 (101)	0.7770	0.7831	0.7827	−0.0005	−4	−56	−0.0093
2 (202)	0.7740	0.7768	0.7789	+0.0021	0	−93	−0.0083
3 (303)	0.7749	0.7801	0.7809	+0.0008	−3	−1	−0.0099
4 (404)	0.7675	0.7778	0.7766	−0.0012	−18	+63	−0.0066

Split (seed)	Raw B	SC B	SCG B	ΔB (SCG – SC)	ΔUFA	ΔBG FP	ΔAP
5 (505)	0.7755	0.7810	0.7784	−0.0026	+4	+63	−0.0086
Mean	0.7738	0.7798	0.7795	−0.0003	−4.2	−4.8	−0.0085

Note. All differences are SCG minus SC. Each split uses 1000 calibration and 4000 test images and refits the ranker, priors and policy. The ranker family and C are kept at their final values; because these were chosen on the original calibration images, which overlap the new test sets, the splits measure sensitivity to refitting rather than to the choice of configuration (see Section S3.6). SC keeps the within-class ranking of Raw, so its full-list AP equals that of Raw in every split.

By geometry group (Table S8). Adding one group of geometry terms at a time to SC shows that box size carries the useful information: size alone reproduces the full UFA reduction of 24 objects, with a smaller AP loss than the full model. Boundary terms reduce UFA by 15 but add background accepts; aspect ratio and position have little effect.

Table S8. Adding each geometry group to SC (COCO, YOLO-World-I).

Added group (terms)	Features	ΔB	ΔUFA	ΔBG FP	ΔAP
Size (width, height, area)	125	+0.0007	−24	+56	−0.0036
Aspect ratio	83	+0.0001	−2	−6	−0.0003
Position (centre x, y)	104	+0.0002	−7	−10	−0.0016
Boundary (border gap, border count)	104	−0.0007	−15	+59	−0.0021
All eight terms (SCG)	230	+0.0001	−24	+15	−0.0057

Note. Differences are relative to SC (62 features). Each group is added together with its class interactions, and the threshold is reselected on calibration images. These results correspond to Fig. 6 of the paper.

By class (Table S9). Geometry reduces the number of accepted detections on unknown objects in 11 of the 20 known classes and increases it only for cow. The largest reductions are for bench, dog and sheep. For bench, recall also rises; for sheep, precision rises and recall falls. AP falls in 11 classes and rises in 9.

Table S9. Per-class effect of geometry (SCG – SC; COCO, YOLO-World-I).

Class	Test objects	Δ unknown-overlap detections	ΔBG FP	ΔR	ΔP	ΔAP
bench	343	−7	+17	+0.0233	−0.0060	+0.0129
dog	174	−4	+1	−0.0172	0.0000	−0.0257
sheep	308	−3	−16	−0.0552	+0.0333	+0.0054
person	8816	−2	+16	+0.0010	−0.0012	−0.0028
train	165	−1	0	0.0000	+0.0104	+0.0009
fire hydrant	80	−1	+2	+0.0250	−0.0096	−0.0297
motorcycle	326	−1	0	−0.0123	−0.0097	−0.0158
car	1574	−1	−4	−0.0032	+0.0041	+0.0045
boat	366	−1	−2	+0.0191	+0.0200	+0.0010
horse	232	−1	−4	+0.0086	+0.0144	−0.0042
airplane	115	−1	−1	+0.0174	+0.0110	−0.0052
truck	318	0	0	+0.0094	+0.0046	−0.0009

Class	Test objects	Δ unknown-overlap detections	Δ BG FP	Δ R	Δ P	Δ AP
stop sign	68	0	-2	0.0000	+0.0207	+0.0063
bus	228	0	-4	-0.0263	+0.0167	-0.0139
bicycle	259	0	-3	-0.0193	+0.0031	+0.0002
traffic light	530	0	+6	+0.0075	-0.0045	-0.0014
cat	162	0	-1	-0.0062	-0.0011	+0.0002
bird	311	0	-2	-0.0096	+0.0027	+0.0034
parking meter	54	0	0	+0.0185	+0.0256	-0.0310
cow	346	+2	+12	-0.0145	-0.0418	-0.0175

Note. "Unknown-overlap detections" counts accepted detections of the given predicted class that overlap an annotated unknown object at $\text{IoU} \geq 0.50$; it is a detection count, so its total (-21) differs from the object-level UFA change (-24). Classes are sorted by this column. Per-class values come from small counts and should be read as descriptive.

S3.3 Comparison with per-class thresholds

Per-class raw thresholds are the strongest simple baseline. Table S10 compares them with GORC-RF on the same test images. RF retains more known objects and has a higher B, whereas per-class thresholds accept fewer unknown objects and fewer background boxes. Neither policy is better on all measures.

Table S10. GORC-RF versus per-class raw thresholds, with 95% paired intervals.

Metric	GORC-RF	Per-class thresholds	Difference (RF – per-class)
B	0.7789	0.7748	+0.0042 [0.0015, 0.0071]
P	0.7487	0.7649	-0.0162 [-0.0212, -0.0109]
R	0.6650	0.6432	+0.0218 [0.0179, 0.0260]
UFA	176	148	+28 [8, 48]
BG FP	2828	2469	+359 [269, 449]
Full-list AP	0.4858	0.4859	-0.0002 [-0.0024, 0.0024]

Note. Intervals use 2000 paired image resamples (500 for AP). Per-class thresholds do not change the ranking of candidates, so their full-list AP equals that of Raw. The per-class thresholds were chosen on calibration images by one coordinate-wise pass over the 20 classes, maximizing B.

S3.4 Role of annotated unknown objects

Full calibration uses annotated unknown objects in three places: the risk weights (Eq. S1), the class priors (Eq. S3) and the B criterion used to choose the policy. To measure their contribution, we repeated fitting and policy selection with known-class labels only. This setting removes all unknown annotations before fitting, uses uniform risk weights and the prior $p_{TP,y}$, and selects the policy by known-class F1. The binary training labels are identical in both settings, because they depend only on known-class annotations. The ranker family and C are the same in both settings; they were chosen in the full-annotation experiment.

Table S11 places the two settings next to the Raw and per-class baselines. Because the two baselines also use B, and hence unknown annotations, to choose their thresholds, this sequence describes the gains rather than isolating each one exactly. Of the B gain of GORC-RF over Raw (+0.0060), about 0.0019 is already obtained by

per-class thresholds, a further 0.0034 by the learned ranker and policy search using known labels only, and 0.0007 by the unknown annotations. The unknown annotations also reduce UFA by 15 objects, at the cost of 37 additional background accepts.

Table S11. Calibration with and without annotated unknown objects.

Method	Uses unknown annotations	B	P	R	UFA	BG FP	Full-list AP
Raw global threshold	Threshold chosen by B	0.7729	0.7384	0.6638	299	2875	0.4859
Per-class raw thresholds	Thresholds chosen by B	0.7748	0.7649	0.6432	148	2469	0.4859
GORC, known-class labels only	No	0.7782	0.7496	0.6632	191	2791	0.4860
GORC-RF, full calibration	Yes	0.7789	0.7487	0.6650	176	2828	0.4858
Full – known-only(95% interval)		+0.0007 [0.00003, 0.00146]			-15 [-24, -7]	+37 [18, 56]	

Note. The two GORC rows use the same images, features, C, score functions and thresholds. Across the five repeated splits of Table S7, full calibration also reduced UFA in every split, while the sign of the B difference varied.

Table S12 separates the three uses of unknown annotations on the primary split. Changing the selection criterion from F1 to B does not change the selected policy, and neither does replacing the full prior with the known-only prior. On this primary split, these controls attribute the selected-policy difference to risk weighting.

Table S12. Effect of risk weights and selection criterion on the selected policy.

Risk weights	Selection criterion	Selected policy	B	UFA	BG FP
With unknown/background weights	F1	Blend $\alpha = 0.8$, $\tau = 0.70$	0.7789	176	2828
With unknown/background weights	B	Blend $\alpha = 0.8$, $\tau = 0.70$	0.7789	176	2828
Uniform	F1	Blend $\alpha = 0.8$, $\tau = 0.70$	0.7782	191	2791
Uniform	B	Blend $\alpha = 0.8$, $\tau = 0.70$	0.7782	191	2791

Note. All four runs use the same class prior so that only the weights and the criterion change.

S3.5 Sensitivity to the known-class vocabulary

For each vocabulary in Table S4, YOLO-World-l was rerun on all 5000 images with the corresponding prompt, and the ranker, priors and policy were refitted on the 1000 calibration images of that stream. The ranker family and C were kept fixed. RF improves B in all five vocabularies, and every interval excludes zero. UFA falls in all five, by 6% to 41%. Background accepts fall only for the primary vocabulary and rise by 3% to 30% in the other four; the largest increase is for VOC20. The random vocabularies contain about one third as many known objects as the primary one and differ in class composition. Both methods reach a lower B on them.

Table S13. Raw and GORC-RF on five known-class vocabularies, with the distinct AP-C policy shown separately (test images).

Vocabulary	B: Raw → policy	ΔB [95% interval]	UFA: Raw → policy	BG FP: Raw → policy	Full-list AP: Raw → policy
Primary (first 20)	0.7729 → 0.7790	+0.0061 [0.0040, 0.0083]	299 → 175 (−41%)	2875 → 2828 (−2%)	0.4860 → 0.4858
VOC20	0.7467 → 0.7561	+0.0094 [0.0060, 0.0125]	397 → 375 (−6%)	2915 → 3800 (+30%)	0.4766 → 0.4774
Random 1 (seed 611)	0.6759 → 0.6952	+0.0193 [0.0114, 0.0274]	594 → 412 (−31%)	755 → 820 (+9%)	0.4563 → 0.4493
Random 2 (seed 612)	0.6472 → 0.6610	+0.0138 [0.0073, 0.0201]	660 → 513 (−22%)	835 → 860 (+3%)	0.4563 → 0.4564
Random 3 (seed 613)	0.6526 → 0.6667	+0.0141 [0.0086, 0.0197]	483 → 350 (−28%)	1212 → 1271 (+5%)	0.3751 → 0.3718
Random 1, AP-C (vs Raw)	0.6759 → 0.6966	+0.0207 [0.0137, 0.0280]	594 → 594 (0%)	755 → 1051 (+39%)	0.4563 → 0.4577

Note. Intervals use 2000 paired image resamples. The primary-vocabulary row comes from a fresh detector run and therefore differs slightly from the main results (RF: B 0.7790 vs 0.7789, UFA 175 vs 176). AP-C selects the same policy as RF in four vocabularies. For random set 1 it selects a different policy, with B 0.6966, full-list AP 0.4577, UFA 594 and BG FP 1051: the AP constraint keeps AP above Raw but removes the UFA reduction and increases background accepts.

S3.6 Dependence on the calibration data

Calibration budget (Table S14). The final ranker family and C were kept fixed, while fitting and policy selection used nested random subsets of 25 to 500 calibration images (three subsets per size). B rises with the number of calibration images, and its variation across subsets becomes small from 100 images onward. UFA and BG FP vary strongly between subsets when fewer than 100 images are used.

Table S14. GORC-RF with smaller calibration sets (mean $\pm$ standard deviation over three subsets).

Calibration images	B	UFA	BG FP	Full-list AP
25	0.7621 $\pm$ 0.0038	193 $\pm$ 123	3032 $\pm$ 1251	0.4738 $\pm$ 0.0046
50	0.7623 $\pm$ 0.0166	182 $\pm$ 142	2640 $\pm$ 1037	0.4793 $\pm$ 0.0055
100	0.7734 $\pm$ 0.0015	159 $\pm$ 29	2936 $\pm$ 523	0.4756 $\pm$ 0.0055
250	0.7765 $\pm$ 0.0017	171 $\pm$ 38	2790 $\pm$ 133	0.4785 $\pm$ 0.0057
500	0.7773 $\pm$ 0.0005	170 $\pm$ 27	2930 $\pm$ 210	0.4806 $\pm$ 0.0042
1000 (full set)	0.7789	176	2828	0.4858

Note. Test results on the same 4000 images. Raw: B 0.7729, UFA 299, BG FP 2875.

Calibration and test performance of the selected policy. On the primary split, the selected RF policy has B = 0.7831 on the calibration images and 0.7789 on the test images; for Raw, the values are 0.7736 and 0.7729. The gain over Raw therefore shrinks from 0.0095 on calibration to 0.0060 on test. The calibration full-list AP of RF exceeds that of Raw by 0.0073, whereas on test images it is 0.0002 lower.

Nested selection (Table S15). To separate selection from fitting completely, each of three runs split the 1000 calibration images into 500 fitting and 500 selection images. The ranker family and C were chosen inside the fitting images (350/150 split), the ranker was refitted on all 500 fitting images, and the policy was chosen on the

500 selection images. We also chose the policy on the fitting images themselves, as in the main protocol, for comparison. All runs were evaluated on the original 4000 test images.

None of the three runs selected the final configuration. With separate selection images, RF improved B over Raw in two runs and was essentially unchanged in the third (mean +0.0025, compared with +0.0060 in the main protocol). When the policy was selected on the fitting images instead, the mean B gain was +0.0016, run 3 lost B, and AP-C lost 0.0228 AP in run 3. These three runs suggest that separate policy-selection images reduce the optimism of the AP gate.

Table S15. Nested configuration, fitting and policy selection (test results).

Run	Configuration selected	Separate selection: RF ΔB	AP-C ΔB	AP-C ΔAP	Selection on fitting images: RF ΔB	AP-C ΔAP
1	Additive, C = 0.1	+0.0041	+0.0041	+0.0005	+0.0053	+0.0007
2	Additive, C = 0.1	-0.00003	0.0000	0.0000	+0.0030	-0.0010
3	Interactions with profiles, C = 1	+0.0035	+0.0035	0.0000	-0.0035	-0.0228

Note. Differences are relative to the Raw policy selected in the same way. In run 2 with separate selection, AP-C retained the Raw policy, so it shows no change in either B or AP. Only three runs were performed, so these results indicate the direction of the effect rather than its size.

S3.7 Alternative selection criteria and background budgets

B weights precision, recall and unknown rejection equally. Table S16 reselects the policy on calibration images under a weighted harmonic mean, or under a limit on the number of background accepts in the calibration images. Doubling the weight on unknown rejection, or applying the background levels of Raw or ordinary Platt scaling as a budget, does not change the selected policy. Doubling the weight on precision, or applying the stricter background level of per-class thresholds, selects the direct ranker with a higher threshold. This roughly halves UFA and reduces BG FP by about 40%, at a recall cost of 0.06–0.07. Doubling the weight on recall selects a lower threshold with more false accepts.

Table S16. Policies selected under alternative calibration criteria (test results).

Calibration criterion	Selected policy	B	P	R	UFA	BG FP
B, weights P:R:URR = 1:1:1 (RF)	Blend $\alpha = 0.8$, $\tau = 0.70$	0.7789	0.7487	0.6650	176	2828
Weights 1:1:2	Blend $\alpha = 0.8$, $\tau = 0.70$	0.7789	0.7487	0.6650	176	2828
Weights 2:1:1	Direct, $\tau = 0.90$	0.7680	0.8181	0.5959	83	1686
Weights 1:2:1	Direct, $\tau = 0.70$	0.7727	0.6945	0.7008	223	3967
B with BG budget of Raw	Blend $\alpha = 0.8$, $\tau = 0.70$	0.7789	0.7487	0.6650	176	2828
B with BG budget of ordinary Platt	Blend $\alpha = 0.8$, $\tau = 0.70$	0.7789	0.7487	0.6650	176	2828
B with BG budget of per-class thresholds	Direct, $\tau = 0.894$	0.7700	0.8123	0.6028	86	1776

Note. A background budget limits the number of background accepts per calibration image to the value achieved by the reference method on the calibration images; the per-class budget corresponds to 519 background accepts on the 1000 calibration images. The Raw and Platt budgets are not binding at the RF policy.

Preference scores (Table S17). Table S17 gives the values plotted in Fig. 5 of the paper. Each score is the weighted sum in Eq. (11) of the paper, with weights $(c_u, c_b, c_m) = (10, 1, 1)$ for the unknown-risk-heavy setting, $(1, 10, 1)$ for the background-risk-heavy setting and $(1, 1, 10)$ for the coverage-sensitive setting. Lower is better. No policy is reselected for this comparison.

Table S17. Preference-weighted risk of the fixed policies (lower is better).

Policy	B	R	UFA	BG FP	Unknown-heavy	Background-heavy	Coverage-sensitive
Raw global threshold	0.7729	0.6638	299	2875	0.7572	2.5211	3.5987
Raw per-class thresholds	0.7748	0.6432	148	2469	0.6567	2.3540	3.7764
Class-balanced logit, fine grid	0.7692	0.6285	231	2232	0.7155	2.2464	3.9168
Class-balanced logit, common grid	0.7731	0.6675	302	2945	0.7586	2.5472	3.5653
Risk-weighted score logistic, C = 0.5	0.7730	0.6627	294	2847	0.7536	2.5088	3.6079
Risk-weighted score logistic, C = 0.1	0.7727	0.6617	292	2836	0.7529	2.5062	3.6182
Ordinary Platt scaling	0.7725	0.6728	315	3090	0.7688	2.6092	3.5193
Temperature scaling	0.7730	0.6665	302	2930	0.7590	2.5423	3.5742
Isotonic regression	0.7725	0.6592	286	2792	0.7495	2.4894	3.6409
GORC-SC	0.7791	0.6654	169	2833	0.6659	2.5029	3.5734
GORC-SG	0.7736	0.6636	235	2960	0.7200	2.5807	3.6027
GORC-SCG	0.7792	0.6646	145	2848	0.6518	2.5170	3.5807
GORC-RF / AP-C	0.7789	0.6650	176	2828	0.6709	2.5018	3.5771

Note. Bold marks the lowest value in each preference column. The class-balanced logit is a score-only logistic model ($C = 0.5$, balanced class weights, no risk weights); "fine grid" uses a threshold grid (22 fixed values, 25 score quantiles and the score extremes), and "common grid" uses the grid of Table S1. The $C = 0.5$ risk-weighted model is included only here as a further control.

S3.8 AP at matched operating points

The AP constraint in AP-C applies to the full candidate list. To compare the accepted sets directly, we kept each score function fixed and chose a new threshold on the calibration images so that each method matched the calibration recall or the calibration accepted count of Raw. Table S18 reports the resulting test values. At calibration-matched accepted counts, RF has similar recall to Raw, accepts 40% fewer unknown objects (177 vs 297) and a similar number of background boxes (2839 vs 2868), while its accepted-set AP is 0.0035 lower. Official COCO evaluation (Table S19) shows the same pattern: RF slightly increases full-list AP and slightly decreases accepted-set AP.

Table S18. Test results at thresholds matched to Raw on the calibration images.

Score function	Matched quantity	Threshold	Test R	Accepted	UFA	BG FP	Full-list AP	Accepted-set AP
Raw	Recall	0.3007	0.6638	13,274	297	2868	0.4859	0.4479
SC	Recall	0.8132	0.6590	12,857	159	2672	0.4859	0.4392
SCG	Recall	0.8212	0.6550	12,707	137	2616	0.4803	0.4328
RF / AP-C	Recall	0.7214	0.6548	12,702	157	2598	0.4858	0.4408
Raw	Accepted count	0.3005	0.6638	13,275	297	2868	0.4859	0.4479

Score function	Matched quantity	Threshold	Test R	Accepted	UFA	BG FP	Full-list AP	Accepted-set AP
SC	Accepted count	0.7987	0.6663	13,170	169	2853	0.4859	0.4436
SCG	Accepted count	0.7997	0.6652	13,126	146	2851	0.4803	0.4373
RF / AP-C	Accepted count	0.6985	0.6657	13,148	177	2839	0.4858	0.4444

Note. Targets are the Raw recall (0.6520) and accepted count (3011) on the calibration images, so the test values match only approximately. These thresholds differ from the B-selected operating points in Table 3 of the paper.

Table S19. Official COCO evaluation (pycocotools) of full candidate lists and accepted sets.

Policy	Full-list AP	Full-list AP50	Full-list AP75	Accepted-set AP	Accepted detections
Raw	0.4941	0.6715	0.5265	0.4534	13,283
GORC-SC	0.4941	0.6715	0.5265	0.4483	13,136
GORC-SG	0.4924	0.6713	0.5270	0.4502	13,284
GORC-SCG	0.4897	0.6700	0.5227	0.4432	13,114
GORC-RF / AP-C	0.4946	0.6727	0.5272	0.4498	13,124

Note. Official COCO evaluation uses the original COCO annotations of the 20 known categories, the standard crowd, area and ignore rules, and at most 100 detections per image. It therefore differs slightly from the full-list AP used in the paper, which keeps every candidate.

S3.9 Dependence on annotation and matching

Stricter matching (Table S20). The accepted detections and thresholds of each policy were kept fixed, and only the IoU required for a match was raised from 0.50 to 0.75. All methods lose recall and gain background accepts, because more detections fail the stricter overlap. UFA falls for the same reason, not because more unknown objects are rejected. SC, SCG and RF all retain higher B than Raw, although their ordering among themselves changes.

Table S20. Fixed policies evaluated at IoU 0.50 and 0.75.

Policy	B at 0.75	UFA at 0.75	BG FP at 0.75	BG FP at 0.50	Median max-IoU of BG FP boxes at 0.50
Raw	0.6955	238	4330	2875	0.040
GORC-SC	0.6999	132	4272	2833	0.038
GORC-SCG	0.7000	114	4278	2848	0.041
GORC-RF / AP-C	0.7003	142	4253	2828	0.039

Note. The last column is the median, over BG FP boxes, of the largest IoU with any known or unknown annotation. For RF, 82.5% of BG FP boxes overlap some annotation, 17.4% reach IoU 0.30 and none reaches 0.50.

Unannotated objects (Table S21). A background false accept is an accepted box whose IoU is below 0.50 with every COCO annotation; the region may still contain an object that COCO does not label. To measure additional annotation support, we compared the BG FP boxes with the LVIS v1 validation annotations. LVIS is an independent human annotation of the same images with 1203 categories, covering 4809 of the 5000 images (3848 of the 4000 test images). About one in six BG FP boxes overlaps a LVIS object at $\text{IoU} \geq 0.50$. The support fractions were similar for Raw (16.2%), RF (15.4%) and per-class thresholds (16.0%), indicating similar rates of additional annotation support in this diagnostic.

Table S21. Annotation support and author inspection of BG FP boxes. (a) Matches to LVIS annotations (IoU ≥ 0.50).

Policy	BG FP	BG FP on LVIS-annotated images	Matching a LVIS box	Fraction [95% interval]
Raw	2875	2792	453	16.2% [13.1%, 20.0%]
GORC-RF / AP-C	2828	2748	423	15.4% [12.4%, 19.1%]
Per-class thresholds	2469	2399	385	16.0% [12.4%, 20.6%]

Note. Intervals use 2000 image resamples. For 406 of the 423 RF boxes, the matching LVIS object has no COCO annotation at IoU ≥ 0.50 . LVIS uses federated annotation, so some objects remain unlabelled. Because LVIS categories and box conventions differ from those of COCO, these fractions measure additional annotation support approximately. The matched boxes are therefore not reassigned to UFA, and all results in the paper use the COCO annotations.

Author inspection. One author inspected a simple random sample of 150 of the 2828 RF background detections on the primary COCO test split. The sample covered 125 images and was drawn without replacement using seed 20261004. The review page showed each box in the full image and in a local crop, without its prediction label, score or LVIS match.

The author classified 112 boxes (74.7%) as recognizable objects, 29 (19.3%) as object parts or ambiguous regions, 3 (2.0%) as background or texture, and 6 (4.0%) as unable to judge (Table S21(b)). Of the 112 recognizable cases, 103 recorded names matched the 20 known classes, including 81 people. Among those 103 cases, 89 boxes were rated as fitting the object well. Nine names (6.0% of the full sample) fell outside the known list: three signs, clothing, wheelchair, hand, chair, rhino and foot. This name-based group includes generic labels and body parts, so it is not an estimate of distinct unknown-object accepts.

Crowd regions. COCO represents some dense groups with crowd annotations. Our evaluator treats their bounding boxes as ordinary annotation boxes. A detection of an individual object within a large crowd box can have IoU below 0.50 and be labelled BG FP. In the sample, 71 of 150 detections have at least half their area inside a crowd annotation box. The calculation takes the largest intersection-area fraction over crowd boxes of any category, using boxes rather than segmentation masks. Across all BG detections, the fractions are 46.6% for RF (1318 of 2828), 45.0% for Raw (1295 of 2875), 46.1% for SC and 46.5% for SCG. Crowd-box overlap is therefore frequent under all four policies. All reported UFA and BG FP metrics retain the original COCO-based counting rules. Online Resource 2 includes the diagnostic script and per-detection results.

Table S21 (continued). (b) Author inspection of the 150 sampled RF background detections.

Visual category	Detection boxes	Percentage of sample
Recognizable object	112	74.7%
Object part or ambiguous region	29	19.3%
Background or texture	3	2.0%
Unable to judge	6	4.0%
Total	150	100.0%

Note. Ratings by one author; percentages refer to all 150 sampled detection boxes.

S3.10 Prompt format (Grounding DINO)

Grounding DINO accepts free-form text, so the prompt format changes the candidate stream. We ran three prompt formats on a fixed subset of 400 COCO images (100 calibration, 300 test) and calibrated each resulting

stream separately. With the period-separated prompt used in the paper, RF changes B only slightly on this small subset. The comma-separated and photo-phrase prompts produce far fewer candidates and very low recall; with such streams, RF lowers B in both cases. The acceptance layer therefore depends on a reasonable candidate stream from the detector.

Table S22. Grounding DINO with three prompt formats (300 test images).

Prompt format	Candidates on all 400 images, before / after suppression	B: Raw → RF	Full-list AP: Raw → RF	UFA: Raw → RF	BG FP: Raw → RF	RF policy
Period-separated ("person. bicycle. ...")	3914 / 2956	0.7477 → 0.7483	0.4712 → 0.4768	0 → 2	262 → 331	Direct, $\tau = 0.50$
Comma-separated ("person, bicycle, ...")	610 / 553	0.0155 → 0.0000	0.0090 → 0.0055	119 → 2	67 → 3	Direct, $\tau = 0.80$
Photo phrases ("a photo of a person", ...)	99 / 96	0.1167 → 0.0278	0.0800 → 0.0769	8 → 0	26 → 6	Blend $\alpha = 0.8$, $\tau = 0.50$

Note. Candidate counts cover all 400 images (100 calibration and 300 test), before and after output-side suppression; the performance columns cover only the 300 test images. The corresponding test-only counts after suppression are 2190, 416 and 82. Photo phrases concatenate each known name as "a photo of a {class}" with " " separators and a final period. On the full 4000 test images, the period-separated prompt gives RF a B gain of +0.0112 (Table 7 of the paper).

S3.11 Processing time

Table S23 times the complete output-side procedure, starting from in-memory boxes, labels, raw scores and image sizes, and ending with the accepted detections. The procedure includes geometry construction, standardization, scoring, score transformation and thresholding. Detector inference, NMS and file input/output are not included. The dense implementation uses the 230-dimensional feature vector; the folded implementation uses the per-class lookup of Eq. (S4). Both give identical accepted sets.

Table S23. Processing time for all 100,497 test candidates (4000 images, one CPU thread).

Mode	Implementation	Median (ms)	95th percentile (ms)	Per image (ms)
SCG	Dense	169.0	173.8	0.0423
RF	Dense	166.3	172.1	0.0416
AP-C	Dense	167.0	172.7	0.0417
SCG	Folded	31.8	33.5	0.0080
RF	Folded	32.1	33.1	0.0080
AP-C	Folded	32.0	33.5	0.0080

Note. Each value is from 100 repetitions after 10 warm-up runs on an Intel Core i7-12800HX. "Per image" divides the batch median by 4000; it is not single-image latency. For single images with representative candidate counts, folded RF takes 0.146–0.179 ms (median). Offline costs, measured separately: fitting the ranker 0.53 s, estimating priors 0.42 s, policy selection over the full grid 0.80 s, and the configuration search of Table S2 51.6 s.